



Fingerprint Management System™ (FMS) Protocol Manual

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Chapter 1. Packet Structure

1.1. Fingerprint Management System Packet Structure

The FMS communication packet is composed of 12 bytes as follows:

Channel	Command	Param1	Param2	Data Size1	Data Size2	Error Code	Check Sum
1 byte	1 byte	2 bytes	2 bytes	2 bytes	2 bytes	1 byte	1 byte

Channel	Communication channel selection. FMS has only one channel. It is always 0x00.																					
Command	Control command to FMS. Refer to 1.3 Command & ACK List.																					
Param1,2	Param1 and Param2 are used to specify the parameters or sub command.																					
	Example: When reading a fingerprint image																					
	<table><tr><td>Channel</td><td>Command</td><td>Param1</td><td>Param2</td><td>Data size1</td><td>Data size2</td><td>Check sum</td></tr><tr><td>0x00</td><td>0x043</td><td>1 or 2</td><td>0x0000</td><td>0x00</td><td>0x00</td><td>0xXX</td></tr></table>	Channel	Command	Param1	Param2	Data size1	Data size2	Check sum	0x00	0x043	1 or 2	0x0000	0x00	0x00	0xXX							
Channel	Command	Param1	Param2	Data size1	Data size2	Check sum																
0x00	0x043	1 or 2	0x0000	0x00	0x00	0xXX																
	Command: CMD_GET_IMAGE (0x43) Param1: 0x0001 (original image size) or 0x0002 (quarter image size) Param1 determines the image size according to the value																					
Data Size1,2	When the Data Size (32bit value) is a non-zero value, additional data packet will be added. Data size1: low extra data size (2 bytes) Data size2: high extra data size (2 bytes)																					
	Example: When registering a fingerprint record to the FMS database																					
	<table><tr><td>Channel</td><td>Command</td><td>Param1 / 2</td><td>Data size1 / 2</td><td>Err Code</td><td>Check sum</td><td>Extra data</td></tr><tr><td>0x00</td><td>0x71</td><td>0x0000 0x0000</td><td>0x0330 0x0000</td><td>0x00</td><td>0x0D</td><td>816 bytes</td></tr></table>	Channel	Command	Param1 / 2	Data size1 / 2	Err Code	Check sum	Extra data	0x00	0x71	0x0000 0x0000	0x0330 0x0000	0x00	0x0D	816 bytes							
Channel	Command	Param1 / 2	Data size1 / 2	Err Code	Check sum	Extra data																
0x00	0x71	0x0000 0x0000	0x0330 0x0000	0x00	0x0D	816 bytes																
	If Extra Data is 816 bytes, Data Size is 816 bytes (0x0000 0330)																					
Error Code	Error code is returned with Acknowledge packet. It shows the result from the previously executed command. Refer to 1.4 Error Code List.																					
Check Sum	Checksum value is the sum of all other 11 bytes, and carry-over is ignored.																					
	Example:																					
	<table><tr><td>Channel</td><td>Command</td><td>Param1</td><td>Param2</td><td>Size 1</td><td>Size 2</td><td>Err Code</td><td>Check sum</td></tr><tr><td>0x00</td><td>0x05</td><td>0x0102</td><td>0x0304</td><td>0x1122</td><td>0x3344</td><td>0x00</td><td>0xB9</td></tr></table>	Channel	Command	Param1	Param2	Size 1	Size 2	Err Code	Check sum	0x00	0x05	0x0102	0x0304	0x1122	0x3344	0x00	0xB9					
Channel	Command	Param1	Param2	Size 1	Size 2	Err Code	Check sum															
0x00	0x05	0x0102	0x0304	0x1122	0x3344	0x00	0xB9															
	Check Sum = 0x00+0x05+0x01+0x02+0x03+0x04+0x11+0x22+0x33+0x44+0x00 = 0xB9																					
Extra Data	Image, DB record, etc. If checksum option of SYSINFO is enabled, add extra data checksum.																					
	Example:																					
	Extra data = 0x11+0x22+0x33+0x44 (when checksum option is disabled) Extra data = 0x11+0x22+0x33+0x44+0xAA (when checksum option is enabled) (Checksum = 0x11+0x22+0x33+0x44 = 0xAA)																					

1.2. Little-Endian Packet Structure

Packets are sent to FMS in little-endian format. Low order bytes precede high order bytes. In the example below, the data structure to be sent to FMS is shown in Table 1. The byte stream that is sent to the FMS is shown in Table 2.

Table 1 – Packet Structure

Channel	Command	Param1	Param2	DataSize1	DataSize2	Error Code	Check Sum
1 byte	1 byte	2 bytes	2 bytes	2 bytes	2 bytes	1 byte	1 byte
0x00	0x05	0x0102	0x0304	0x1122	0x3344	0x00	0xB9

Table 2 – Byte Stream

1	2	3	4	5	6	7	8	9	10	11	12
0x00	0x05	0x02	0x01	0x04	0x03	0x22	0x11	0x44	0x33	0x00	0xB9

1.3. Command and ACK List

Code	Command & ACK	Description
0x05	CMD_GET_VERSION	Read firmware version information
0x12	***CMD_RELAY_ON_OFF	Control Relay0 and Relay1 out signals
0x15	***CMD_SET_WIEGAND_OUT	Output data to Wiegand port
0x19	CMD_DIFF_FP_REGISTER	One time allow two different fingerprints during registration
0x20	**CMD_SET_SYSTEM_INFO	Set FMS system configuration values
0x21	CMD_SET_COMM_SPEED	Set serial communication speed
0x30	CMD_GET_SYSTEM_INFO	Read FMS system configuration values
0x31	CMD_SET_EXT_USER_ID	Set Extended User ID
0x32	CMD_GET_EXT_USER_ID	Get Extended User ID
0x35	CMD_MAKE_RECORD_START	Initiate process of generating multiple-view SG400, ANSI378 or ISO19794-2 fingerprint records
0x36	CMD_MAKE_RECORD_CONT	Capture a fingerprint image and extract template. Templates accumulate in RAM until CMD_MAKE_RECORD_END is received.
0x37	CMD_MAKE_RECORD_END	Finalize and download SG400, ANSI378 or ISO19794-2 template

Code	Command & ACK	Description
0x40	CMD_GET_TEMPLATE	Read fingerprint template data of input fingerprint image from the sensor
0x43	CMD_GET_IMAGE	Read fingerprint image data from the sensor
0x46	CMD_GET_REC_COUNT	Read the number of registered users
0x47	CMD_GET_MASTER_COUNT	Read the number of registered masters
0x50	**CMD_FP_REGISTER_START	These commands are used to register a fingerprint. The first fingerprint is captured by the CMD_FP_REGISTER_START command, and the second fingerprint is captured by the CMD_FP_REGISTER_END command.
0x51	**CMD_FP_REGISTER_END	
0x52	**CMD_FP_CHANGE_START	These commands are used to change previously registered fingerprint data. The first fingerprint is captured by the CMD_FP_CHANGE_START command, and the second fingerprint is captured by the CMD_FP_CHANGE_END command.
0x53	**CMD_FP_CHANGE_END	
0x54	**CMD_FP_DELETE	Delete a specified user record from the FMS database
0x55	*CMD_FP_VERIFY	Verify the input fingerprint from the sensor with a registered user in the FMS database
0x56	*CMD_FP_IDENTIFY	Identify the input fingerprint from the sensor with all registered users in the FMS database
0x57	CMD_FP_VERIFY_MASTER	Verify the master before executing a command that requires master authentication. These commands are used when SI_USE_MASTER_AUTHENTICATION option is set.
0x58	CMD_FP_VERIFY_MASTER_END	
0x59	CMD_FP_IDENTIFY_EX	Identify an input fingerprint within the limited range of DB
0x60	CMD_IS_REGISTERED_USER	Check whether a specified user is registered
0x6F	CMD_GET_AUTO_ON	Check Auto On status of sensor module
0x70	CMD_DB_GET_RECCOUNT	Get the number of registered users in the FMS
0x71	**CMD_DB_ADD_REC	Register a fingerprint record to the FMS database from external (host) controller
0x72	**CMD_DB_DELETE_REC	Delete a specified user record from the FMS database
0x73	**CMD_DB_GET_REC	Get a specified user record from the FMS database
0x74	**CMD_DB_GET_FIRSTREC	Get the first user record from the FMS database

Code	Command & ACK	Description
0x75	**CMD_DB_GET_NEXTREC	Get the next user record from the FMS database
0x76	**CMD_DB_DELETE_ALL	Delete all user records in the FMS
0x77	**CMD_DB_GET_CURRENTREC	Get the last record obtained by the commands CMD_DB_GET_FIRSTREC or CMD_DB_GET_NEXTREC
0x78	*CMD_DB_VERIFY	Verify a registered fingerprint in FMS database with a received fingerprint from external (host) controller
0x79	*CMD_DB_IDENTIFY	Identify all registered fingerprints in FMS database with a fingerprint received from external (host) controller
0x7A	CMD_DB_IDENTIFY_EX	Identify an input fingerprint template within the limited range of database
0x7B	CMD_DB_VERIFY_MASTER	Verify the master before executing a command that requires master authentication. These commands are used when SI_USE_MASTER_AUTHENTICATION option is set.
0x7C	CMD_DB_VERIFY_MASTER_END	
0x7D	CMD_DB_GET_ID_LIST	Get all User IDs in FMS database
0x7E	CMD_DB_GET_MASTER_LIST	Get all Master IDs in FMS database
0x92	CMD_SET_USERDATA	Write user data in the flash memory
0x93	CMD_GET_USERDATA	Read user data in the flash memory
0xA1	*CMD_FP_AUTO_IDENTIFY_START	When the CMD_FP_AUTO_IDENTIFY command is received, FMS continuously identifies the input fingerprint from the sensor with all the registered fingerprints in the FMS database until CMD_FP_AUTO_IDENTIFY_STOP command is received. Identification result is sent to external (host) controller whenever a user places a finger on the sensor. When this command is executed, some commands are ignored.
0xA2	CMD_FP_AUTO_IDENTIFY_STOP	
0xA3	*CMD_FP_AUTO_IDENTIFY	Identify the input fingerprint from the sensor with all registered users in the FMS database
0xB0	***CMD_SET_IO_CONFIG	Configure IO
0xB1	***CMD_GET_IO_CONFIG	Get IO configuration
0xB2	***CMD_WRITE_IO	Set OUT signals of IO
0xB3	***CMD_READ_IO	Get IN signals of IO

Code	Command & ACK	Description
0xB6	CMD_GET_VOID_ID_LIST	Get a list of User IDs detected as forged or corrupted
0xD0	CMD_INSTANT_VERIFY	Verify the input fingerprint from the sensor with the received fingerprint from external (host) controller
0xF0	CMD_REQUEST_CONNECT	Check the communication between the FMS and external (host) controller
0xF9	CMD_GET_MATCH_SCORE	Get the user ID and matching score that was generated by the last matching activity from FMS.
0xFA	CMD_LOAD_MATCH_TEMPLATE	Verify two fingerprint templates

* When using these commands, Relay0 or Wiegand signals are generated depending on whether SI_USE_RELAY or SI_WIEGAND_FORMAT options are selected. Relay0 or Wiegand signals are not supported by Bluetooth module.

** These commands are required for master authentication if SI_USE_MASTER_AUTHENTICATION option is selected.

*** Unity 20 Bluetooth does not support Wiegand and GPIO control commands.

1.4. Error Code List

Code	ACK	Description
0x00	ERR_NONE	Normal operation
0x01	ERR_FLASH_OPEN	Flash memory error
0x02	ERR_SENSOR_OPEN	Sensor initialization failed
0x03	ERR_REGISTER_FAILED	Fingerprint registration failed
0x04	ERR_VERIFY_FAILED	Fingerprint verification failed
0x05	ERR_ALREADY_REGISTERED_USER	User ID already exists
0x06	ERR_USER_NOT_FOUND	User ID was not found
0x08	ERR_TIME_OUT	Failed to capture fingerprint in preset time
0x09	ERR_DB_FULL	FMS database is full
0x0A	ERR_WRONG_USERID	Wrong user ID
0x0B	ERR_DB_NO_DATA	FMS database is empty
0x10	ERR_FUNCTION_FAIL	Wrong usage of command packet

Code	ACK	Description
0x11	ERR_INSUFFICIENT_DATA	Wrong length value of Extra Data
0x12	ERR_FLASH_WRITE_ERROR	Flash write error
0x13	ERR_FLASH_READ_ERROR	Flash read error
0x14	ERR_INVALID_PARAM	Parameter value is invalid
0x17	ERR_AUTHENTICATION_FAIL	Master identification failed or needs master authentication
0x1B	ERR_IDENTIFY_FAILED	Fingerprint identification failed
0x28	ERR_CHECKSUM_ERR	Wrong checksum
0x30	ERR_INVALID_FPRECORD	Record format is invalid
0x32	*ERR_DB_MINEX_TO_SIMPLIFIED	Algorithm mismatch. The template database being added was generated by MINEX algorithm, but the device uses SIMPLIFIED algorithm.
0x33	*ERR_DB_SIMPLIFIED_TO_MINEX	Algorithm mismatch. The template database being added was generated by SIMPLIFIED algorithm, but the device uses MINEX algorithm.
0xFE	ERR_BREAK	Image capture was stopped
0xFF	ERR_UNKNOWN_COMMAND	Unknown command

* Related protocol: CMD_DB_ADD_REC. Refer to the PDF document “Compatibility Guide MINEX_SIMPLIFIED algorithms” for more information.

1.5. Warning Code List

Code	ACK	Description
0x41	WARNING_DB_LOADING	Returns when FMS is still in booting process. Refer to “FMS booting time” below.
0x42	WARNING_DB_CHECKSUM	Returns when there are one or more user data that has been detected as forged or corrupted while loading from flash memory.

- FMS booting time
 - FMS booting time depends on the number of users registered in FMS because all user data are required to be loaded into memory upon power up.
 - It takes less than one second by default and one additional second for every 1,000 users.
- WARNING_DB_LOADING
 - This may be returned from FMS when calling the command, CMD_REQUEST_CONNECT (0xF0).
 - To connect and communicate with FMS properly, the host controller must receive ACK with error code 0x00 after sending the command, CMD_REQUEST_CONNECT (0xF0).
 - While loading user data, FMS returns the warning code, WARNING_DB_LOADING (0x41) and ignores all further commands. The host controller must repeat sending the command, CMD_REQUEST_CONNECT (0xF0), until FMS completes data loading and returns ERR_NONE.
- WARNING_DB_CHECKSUM
 - This may be returned from FMS when calling the command, CMD_REQUEST_CONNECT (0xF0).
 - After powering up, FMS automatically loads all user data. If there is any data detected as forged or corrupted while loading and initializing, FMS returns this warning code.
 - In case FMS returns WARNING_DB_CHECKSUM (0x42), it is recommended that the host controller check the void ID list (CMD_GET_VOID_ID_LIST, 0xB6). All users in the list are detected as forged or corrupted and removed from the FMS database so reenrollment might be necessary. Although returning this code, FMS would work normally for all users except for the ones in the void ID list.

1.6. Extra Data Transfer

1.6.1. Extended ID Transfer

- **Example:** Sending extra data for extended user ID (1234567890)
- Data Size 1: 0x0005 / ID: 1234567890

1	2	3	4	5
1 byte	1 byte	1 byte	1 byte	1 byte
0x12	0x34	0x56	0x78	0x90

- **Example:** Sending extra data for extended user ID (1234567890123456789012)
- Data size 1: 0x000B / ID: 1234567890123456789012

1	2	3	4	5	6	7	8	9	10	11
1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte
0x12	0x34	0x56	0x78	0x90	0x12	0x34	0x56	0x78	0x90	0x12

1.6.2. User Record Transfer

- **Example:** Sending a user record with 4-digit ID (1234) length (**SG400 format**)
- Set Extended ID length to 0x00

1	2	3	4	5 ~ 404	405 ~ 804	805 ~ 815	816
User ID		Master Flag		Template1	Template2	Extended ID	Extended ID length
1 byte	1 byte	1 byte	1 byte	400 bytes	400 bytes	11 bytes	1 byte
0x12	0x34	0x00	0x0X	0xFF	0xFF	0xFF	0x00

- **Example:** Sending a user record with 4-digit ID (1234) length (**ANSI378 or ISO19794-2 format**)
- Even if ANSI378 template size is smaller than 800 bytes, template buffer size should be 800 bytes.
- Set Extended ID length to 0x00

1	2	3	4	5 ~ 804	805 ~ 1604	1605 ~ 1615	1616
User ID		Master Flag		Template1	Template2	Extended ID	Extended ID length
1 byte	1 byte	1 byte	1 byte	800 bytes	800 bytes	11 bytes	1 byte
0x12	0x34	0x00	0x0X	0xFF	0xFF	0xFF	0x00

- **Example:** Sending a user record with 10-digit ID (1234567890) (**SG400 format**)
- Set Extended ID length to 0x05
- Set User ID to 0xFFFF

1	2	3	4	5 ~ 404	405 ~ 804	805	806
User ID		Master Flag		Template1	Template2	Extended ID	Extended ID
1 byte	1 byte	1 byte	1 byte	400 bytes	400 bytes	1 byte	1 byte
0xFF	0xFF	0x00	0x0X	0xFF	0xFF	0x12	0x34

807	808	809	810~815	816
Extended ID	Extended ID	Extended ID	Extended ID	Extended ID length
1 byte	1 byte	1 byte	1 byte	1 byte
0x56	0x78	0x90	0xFF	0x05

- **Example:** Sending a user record with 22-digit ID (1234567890123456789012) (**SG400 format**)
- Set Extended ID length to 0x0B
- Set User ID to 0xFFFF
- **In case of ANSI378 or ISO19794-2, template size is 800 bytes.**

1	2	3	4	5 ~ 404	405 ~ 804	805	806
User ID		Master Flag		Template1	Template2	Extended ID	Extended ID
1 byte	1 byte	1 byte	1 byte	400 bytes	400 bytes	1 byte	1 byte
0xFF	0xFF	0x00	0x0X	0xFF	0xFF	0x12	0x34

807	808	809	810	811	812	813
Extended ID	Extended ID	Extended ID	Extended ID	Extended ID	Extended ID	Extended ID
1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte
0x56	0x78	0x90	0x12	0x34	0x56	0x78

814	815	816
Extended ID	Extended ID	Extended ID length
1 byte	1 byte	1 byte
0x90	0x12	0x0B

Chapter 2. Commands

2.1. Setting FMS Hardware Configuration

The FMS hardware and operational settings, including communications speed, template format and various others, are set using SYSINFO commands.

Command : CMD_SET_COMM_SPEED (0x21)
 : CMD_SET_SYSTEM_INFO (0x20)
 : CMD_GET_SYSTEM_INFO (0x30)

2.1.1. SYSINFO Command Table

- Default and available settings are shown in the table below.
- Commands that only accept a zero value for parameter 2 ignore this parameter.

Command Code: SET_SYSTEM_INFO(0x20) or GET_SYSTEM_INFO(0x30)			
Parameter1		Parameter2	
		Default	Setting range
0x00	SI_USE_MASTER_AUTHENTICATION	0	0 or 1
0x02	SI_VERIFY_SECURITY_LEVEL	5	1 ~ 9 Refer to 2.1.3 Security Level
0x03	SI_USE_RELAY	0	0 or 1
0x04	SI_COM_SPEED	0x17 (9600 bps)	Refer to 2.1.4 Communication Speed
0x05	SI_BRIGHTNESS	50	0 ~ 100
0x08	SI_IDENTIFY_SECURITY_LEVEL	5	1 ~ 9 Refer to 2.1.3 Security Level
0x09	SI_WIEGAND_FORMAT	0x012C	0x0000 (not used) 0x012C (26Bit format) 0x012E (34Bit format)
0x0A	SI_WIEGAND_SITE_CODE	0	
0x0B	SI_REGISTER_QUALITY	40	10 ~ 100
0x0C	SI_VERIFY_QUALITY	30	10 ~ 100
0x0E	SI_RELAY_TIME	0	0 ~ 20 (seconds)
0x10	SI_USING_EXTRA_DATA_CHECKSUM	0	0 or 1
0x11	SI_USE_AUTO_ON	1	0 or 1

0x12	SI_SMART_CAPTURE	1	0 or 1
0x13	SI_USING_DIFF_FP_REGISTER	0	0 or 1
0x16	SI_ID_DIGIT	4	4 ~ 22
0x17	SI_SENSOR_TIME_OUT	5	1 ~ 20 (seconds)
0x18	SI_TEMPLATE_TYPE	0x0200	0x0200 (SG400) 0x0100 (ANSI378) 0x0300 (ISO19794-2)

2.1.2. SYSINFO Command Description

Command	Description
SI_USE_MASTER_AUTHENTICATION	Set using master authentication
SI_VERIFY_SECURITY_LEVEL	Set verify security level
SI_USE_RELAY	Set using relay
SI_COM_SPEED	Set serial communication speed
SI_BRIGHTNESS	Set sensor brightness
SI_IDENTIFY_SECURITY_LEVEL	Set identify security level
SI_WIEGAND_FORMAT	Set Wiegand output format
SI_WIEGAND_SITE_CODE	Set Wiegand site code
SI_REGISTER_QUALITY	Set fingerprint quality value for registration
SI_VERIFY_QUALITY	Set fingerprint quality value for verification
SI_RELAY_TIME	Set out signal time of relay
SI_USE_EXTRA_DATA_CHECKSUM	Use check sum of Extra Data
SI_USE_AUTO_ON	Use Auto-On™ function of sensor
SI_SMART_CAPTURE	Use Smart Capture™ for dry or difficult fingerprints
SI_USING_DIFF_FP_REGISTER	Allow different two fingerprints registration
SI_ID_DIGIT	Set extended ID digits
SI_SENSOR_TIME_OUT	Set sensor timeout value. FMS captures fingerprint during this time.
SI_TEMPLATE_TYPE	Set template format (SG400, ANSI378, or ISO19794-2)

2.1.3. Security Level (Verify, Identify)

Security Levels		
1	0x01	SLEVEL_LOWEST
2	0x02	SLEVEL_LOWER
3	0x03	SLEVEL_LOW
4	0x04	SLEVEL_BELOW_NORMAL
5	0x05	SLEVEL_NORMAL
6	0x06	SLEVEL_ABOVE_NORMAL
7	0x07	SLEVEL_HIGH
8	0x08	SLEVEL_HIGHER
9	0x09	SLEVEL_HIGHEST

2.1.4. Communication Speed

COM speed	
0xF3	921,600 bps
0xF2	460,800 bps
0xF1	230,400 bps
0x01	115,200 bps
0x03	57,600 bps
0x05	38,400 bps
0x0B	19,200 bps
0x17	9,600 bps
0x5F	2,400 bps
0xBF	1,200 bps

CMD_SET_COMM_SPEED (0x21)

- Default baud rate is 9600 bps after FMS power-up or reset. During operation, the external (host) controller can change the baud rate, which occurs after ACK packet is returned.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x21	0x21
Param1	Communication Speed	0x0004
Param2	0x0000	Communication Speed
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

CMD_SET_SYSTEM_INFO (0x20)

- Set FMS system configuration values

	COMMAND	ACK
Channel	0x00	0x00
Command	0x20	0x20
Param1	Parameter1	Parameter1
Param2	Parameter2	Parameter2
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

CMD_GET_SYSTEM_INFO (0x30)

- Get FMS system configuration values

	COMMAND	ACK
Channel	0x00	0x00
Command	0x30	0x30
Param1	Parameter1	Parameter1
Param2	0x0000	Return Value
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

CMD_SET_EXT_USER_ID (0x31)

- FDA02 compatible

	COMMAND	ACK
Channel	0x00	0x00
Command	0x31	0x31
Param1	0x0000	0x0000
Param2	0x0000	0x0000
Data size 1	Low Extra Data Size	0x0000
Data size 2	High Extra Data Size	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data	Extended User ID	

CMD_GET_EXT_USER_ID (0x32)

- FDA02 compatible

	COMMAND	ACK
Channel	0x00	0x00
Command	0x32	0x32
Param1	0x0000	0x0000
Param2	0x0000	0x0000
Data size 1	0x0000	Low Extra Data Size
Data size 2	0x0000	High Extra Data Size
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data		Extended User ID

2.2. Fingerprint Registration, Verification, Change and Deletion

2.2.1. Structure of Fingerprint Record

The FMS supports three template formats. The **SG400 format** is SecuGen's proprietary format and is compatible with all SecuGen SDKs that support 400-byte templates. The **ANSI378 format** is compliant with the ANCI-INCITS 378-2004 standard for minutiae template data. The **ISO19794-2 format** is compliant with the ISO/IEC 19794-2 standard "Information technology–Biometric data interchange formats–Part 2: Finger minutiae data. The formats are shown below.

SG400 Format			ANSI378 or ISO19794-2 Format		
Item	Size (bytes)	Description	Item	Size (bytes)	Description
User ID	2	User ID 4 digit	User ID	2	User ID 4 digit
Master Flag	2	0: user 1: master	Master Flag	2	0: user 1: master
Template1	400	SG400 template 1	Template1	800	ANSI or ISO template 1
Template2	400	SG400 template 2	Template2	800	ANSI or ISO template 2
Extended User ID	11	Extended ID 22 digit	Extended User ID	11	Extended ID 22 digit
Extended User ID length	1	Extended ID length	Extended User ID length	1	Extended ID length
<pre>typedef struct _tagUSER_RECORD { U16 id; U16 admin; U8 min1[400]; U8 min2[400]; U8 xid[11]; U8 xid_length; } USER_RECORD;</pre>			<pre>typedef struct tagANSI_RECORD { U16 id; U16 admin; U8 min1[800]; U8 min2[800]; U8 xid[11]; U8 xid_length; } ANSI_RECORD;</pre>		

- FMS supports a maximum of 4-digit standard user ID or 22-digit extended user ID.
- The number of Extended User ID digits must be an even number. Only numerals are allowed for each digit (0 ~ 9).
- User ID range: 000 ~ 999
- Extended User ID range: 00 00000 00000 00000 00000 ~ 99 99999 99999 99999 99999

2.2.2. Registration

- FMS supports two methods to register a fingerprint.
 - CMD_FP_REGISTER_START and CMD_FP_REGISTER_END to register a fingerprint from the sensor
 - CMD_DB_ADD_REC to register a fingerprint from the external (host) controller

Command : CMD_FP_REGISTER_START (0x50)
 : CMD_FP_REGISTER_END (0x51)
 : CMD_DB_ADD_REC (0x71)

Registering Fingerprints Captured from Sensor (CMD_FP_REGISTER*)

- If any error is received during the registration process, the whole registration process should be started again.

CMD_FP_REGISTER_START (0x50)

- Capture the first fingerprint from the sensor.

Example 1: Using standard 4-digit User ID

	COMMAND	ACK
Channel	0x00	0x00
Command	0x50	0x50
Param1	User ID (4 digit)	User ID (4 digit)
Param2	0: user 1: master	0x0000 or 0x0001
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

Example 2: Using extended 6 ~ 22-digit user ID.

- Must set 0xFFFF in the Param1 field in command packet

	COMMAND	ACK
Channel	0x00	0x00
Command	0x50	0x50
Param1	0xFFFF	0xFFFF
Param2	0: user 1: master	0x0000 or 0x0001
Data size 1	Low Extra Data Size	0x0000
Data size 2	High Extra Data Size	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data	Extended User ID	

** [Refer to 1.6.1 Extended ID Transfer](#) (Extra Data)

CMD_FP_REGISTER_END (0x51)

- Capture the second fingerprint from the sensor and end the registration process.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x51	0x51
Param1	0x0000	0x0000
Param2	0x0000	0x0000
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

CMD_FP_DIFF_REGISTER (0x19)

- When a user registers two different fingerprints for one User ID, this command should be sent to the FMS first.
- This command applies only one time to the next user registration.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x019	0x019
Param1	0x0000	0x0000
Param2	0x0000	0x0000
Data Size 1	0x0000	0x0000
Data Size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

Registering Fingerprint Templates from Host Controller (CMD_DB_ADD_REC)**CMD_DB_ADD_REC (0x71)**

- Extra data is user record.
- Fingerprint is registered from external (host) device.
- Template format of data sent to FMS should be the same format as in SYSINFO.
- SG400 template size is 400 bytes. ANSI378 and ISO19794-2 template size is 800 bytes.

Example 1: Using standard 4-digit user ID.

- Must set 0x00 in the Extended ID Length field in command packet.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x71	0x71
Param1	0: do not over write 1: over write	0x0000 or 0x0001
Param2	0x0000	0x0000
Data size 1	Low Extra Data Size	0x0000
Data size 2	High Extra Data Size	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data	User ID (2 bytes)	

	Master Flag (2 bytes)
	Template1 (400 or 800 bytes)
	Template2 (400 or 800 bytes)
	Extended User ID (MAX:11 bytes)
	0x00: Extended ID Length (1 byte)

** [Refer to 1.6.2 User Record Transfer](#) (Extra Data)

Example 2: Using extended 6 ~ 22-digit User ID.

- Must set 0xFFFF in the User ID field in command packet.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x71	0x71
Param1	0: do not over write 1: over write	0x0000 or 0x0001
Param2	0x0000	0x0000
Data size 1	Low Extra Data Size	0x0000
Data size 2	High Extra Data Size	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data	0xFFFF: User ID (2 bytes)	
	Master Flag (2 bytes)	
	Template1 (400 or 800 bytes)	
	Template2 (400 or 800 bytes)	
	Extended User ID (MAX:11 bytes)	
	Extended User ID Length (1 byte)	

** [Refer to 1.6.2 User Record Transfer](#) (Extra Data)

Changing Fingerprint Data

- Use these commands to change previously registered fingerprint data.
- If there is any Error code received during the change process, the whole change process should be started again.

Command : CMD_FP_CHANGE_START (0x52)
: CMD_FP_CHANGE_END (0x53)

CMD_FP_CHANGE_START (0x52)

- The first fingerprint is captured from the sensor.

Example 1: Using standard 4-digit User ID.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x52	0x52
Param1	User ID (4 digit)	User ID (4 digit)
Param2	0: user 1: master	0x0000 or 0x0001
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

Example 2: Using extended 6 ~ 22-digit User ID.

- Must set 0xFFFF in Param1 in command packet.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x52	0x52
Param1	0xFFFF	0xFFFF
Param2	0: user 1: master	0x0000 or 0x0001
Data size 1	Low Extra Data Size	0x0000
Data size 2	High Extra Data Size	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data	Extended User ID	

** [Refer to 1.6.1 Extended ID Transfer](#) (Extra Data)

CMD_FP_CHANGE_END (0x53)

- The second fingerprint is captured from the sensor, and the change process ends.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x53	0x53
Param1	0x0000	0x0000
Param2	0x0000	0x0000
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

Matching and Identifying Fingerprint Data

- For both 1:1 mode (verification) and 1:N mode (identification), the FMS fingerprint algorithms perform matching functions to compare fingerprints captured from the sensor and fingerprint templates stored in the FMS database. The FMS also provides 1:1 verification between a fingerprint captured from the sensor and a template sent from the external (host) controller.

	FMS Fingerprint DB	Host Fingerprint DB
Capture Fingerprint from Sensor	1:1 or 1:N Matching (0x55) CMD_FP_VERIFY (0x56) CMD_FP_IDENTIFY (0x59) CMD_FP_IDENTIFY_EX (0xA3) CMD_FP_AUTO_IDENTIFY (0xA1) CMD_FP_AUTO_IDENTIFY_START (0xA2) CMD_FP_AUTO_IDENTIFY_STOP (0x57) CMD_FP_VERIFY_MASTER (0x58) CMD_FP_VERIFY_MASTER_END	1:1 Matching (0xD0) CMD_INSTANT_VERIFY
Host DB Fingerprint	1:1 or 1:N Matching (0x78) CMD_DB_VERIFY (0x79) CMD_DB_IDENTIFY (0x7A) CMD_DB_IDENTIFY_EX (0x7B) CMD_FP_VERIFY_MASTER (0x7C) CMD_FP_VERIFY_MASTER_END	(0xFA) CMD_LOAD_MATCH_TEMPLATE

CMD_FP_VERIFY_MASTER (0x57)

- Verify master fingerprint by comparing the fingerprint captured from the sensor to the fingerprint templates in the FMS database.
- Master ID is returned in Param1 in ACK packet.

Example 1: Using standard 4-digit User ID.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x57	0x57
Param1	0 or 1 or n	User ID (4 digit)
Param2	0x0000	0x0000
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

Param1	Description
0	All commands after this command will be authenticated by master identification.
1	Only one command after this command will be authenticated.
N	Specify the number of commands to be authenticated (up to 65,535).

Example 2: Using extended 6 ~ 22-digit User ID.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x57	0x57
Param1	0 or 1 or n	0xFFFF
Param2	0x0000	0x0000
Data size 1	0x0000	Low Extra Data Size
Data size 2	0x0000	High Extra Data Size
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data		Extended User ID

CMD_DB_VERIFY_MASTER (0x7B)

- Verify master fingerprint by comparing fingerprint template from external (host) controller to fingerprint templates in the FMS database.
- Template format of data sent to FMS should be the same format as in SYSINFO.
- Master ID is returned in Param1 in ACK packet.

Example 1: Using standard 4-digit User ID.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x7B	0x7B
Param1	0 or 1 or n	User ID (4 digit)
Param2	0x0000	0x0000
Data size 1	Low Extra Data Size	0x0000
Data size 2	High Extra Data Size	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data	SG400, ANSI378 or ISO19794-2 Template	

Param1	Description
0	All commands after this command will be authenticated by master identification.
1	Only one command after this command will be authenticated.
N	Specify the number of commands to be authenticated (up to 65,535).

Example 2: Using extended 6 ~ 22-digit User ID.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x7B	0x7B
Param1	0 or 1 or n	0xFFFF
Param2	0x0000	0x0000
Data size 1	Low Extra Data Size	Low Extra Data Size
Data size 2	High Extra Data Size	High Extra Data Size
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data	SG400, ANSI378 or ISO19794-2 Template	Extended User ID

CMD_FP_VERIFY_MASTER_END (0x58), CMD_DB_VERIFY_MASTER_END (0x7C)

- End master identification process.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x58 or 0x7C	0x58 or 0x7C
Param1	0x0000	0x0000
Param2	0x0000	0x0000
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

CMD_FP_VERIFY (0x55)

- Verify fingerprint by comparing fingerprint captured from the sensor to the fingerprint templates in the FMS database.
- Wiegand or Relay out is generated when verify success.
- Wiegand output is valid when User ID is 4 digits.

Example 1: Using standard 4-digit User ID.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x55	0x55
Param1	User ID (4 digit)	User ID (4 digit)
Param2	* bit 0 0: Do not use Wiegand or Relay0 out 1: Use Wiegand or Relay0 out * bit 1 1: If FMS fails to match, FMS will automatically try again one more time.	matched score
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

Example 2: Using extended 6 ~ 22-digit User ID.

- Must set 0xFFFF in Param1 in command packet.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x55	0x55
Param1	0xFFFF	0xffff
Param2	* bit 0 0: Do not use Wiegand or Relay0 out 1: Use Wiegand or Relay0 out * bit 1 1: If FMS fails to match, FMS will automatically try again one more time.	matched score
Data size 1	Low Extra Data Size	0x0000
Data size 2	High Extra Data Size	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data	Extended User ID	

** [Refer to 1.6.1 Extended ID Transfer](#) (Extra Data)

CMD_FP_IDENTIFY (0x56)

CMD_FP_AUTO_IDENTIFY (0xA3)

- Identify fingerprint by comparing the fingerprint template captured from the sensor to the fingerprint templates in the FMS database.
- Wiegand or Relay out is generated when verify success.
- Wiegand output is valid when User ID is 4 digits.

Example 1: Using standard 4-digit User ID.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x56 or 0xA3	0x56 or 0xA3
Param1	0x0000	User ID (4 digit)
Param2	* bit 0 0: Do not use Wiegand or Relay0 out 1: Use Wiegand or Relay0 out * bit 1 1: If FMS fails to match, FMS will automatically try again one more time.	matched score
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

Example 2: Using extended 6 ~ 22-digit User ID.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x56 or 0xA3	0x56 or 0xA3fdpa
Param1	0x0000	0xFFFF
Param2	* bit 0 0: Do not use Wiegand or Relay0 out 1: Use Wiegand or Relay0 out * bit 1 1: If FMS fails to match, FMS will automatically try again one more time.	matched score
Data size 1	0x0000	Low Extra Data Size
Data size 2	0x0000	High Extra Data Size
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data		Extended User ID

CMD_FP_IDENTIFY_EX (0x59)

- Identify fingerprint by comparing the fingerprint template captured from the sensor within a restricted range of the database.
- Wiegand or Relay out is not generated.
- This command does not work with Extended ID.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x59	0x59
Param1	First ID number of search	User ID (4 digit)
Param2	Last ID number of search	Matched score
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

CMD_DB_VERIFY (0x78)

- Verify fingerprint by comparing the fingerprint template from external (host) controller and the fingerprint template in the FMS database.
- Template format of data sent to FMS should be the same format as in SYSINFO.
- SG400 template size is 400 bytes. ANSI378 and ISO19794-2 template size is 800 bytes.
- Wiegand or Relay out is generated when verify success.
- Wiegand output is valid when User ID is 4 digits.

Example 1: Using standard 4-digit User ID.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x78	0x78
Param1	User ID (4 digit)	User ID (4 digit)
Param2	0: do not use Wiegand or Relay0 out 1: use Wiegand or Relay0 out	matched score
Data size 1	Low Extra Data Size	0x0000
Data size 2	High Extra Data Size	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data	SG400, ANSI378 or ISO19794-2 Template	

Example 2: Using extended 6 ~ 22-digit User ID.

- Must set 0xFFFF in Param1 in command packet.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x78	0x78
Param1	0xFFFF	0xFFFF
Param2	0: do not use Wiegand or Relay0 out 1: use Wiegand or Relay0 out	matched score
Data size 1	Low Extra Data Size	0x0000
Data size 2	High Extra Data Size	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data	Extended User ID (11 bytes) + ID length (1 byte) + SG400, ANSI378 or ISO19794-2 Template	

CMD_DB_IDENTIFY (0x79)

- Identify fingerprint by comparing the fingerprint template from external (host) controller to the fingerprint templates in the FMS database.
- Template format of data sent to FMS should be the same format as in SYSINFO.
- SG400 template size is 400 bytes. ANSI378 and ISO19794-2 template size is 800 bytes.
- Wiegand or Relay out is generated when verify success.
- Wiegand output is valid when User ID is 4 digits.

Example 1: Using standard 4-digit User ID.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x79	0x79
Param1	0x0000	User ID (4 digit)
Param2	0: do not use Wiegand or Relay0 out 1: use Wiegand or Relay0 out	matched score
Data size 1	Low Extra Data Size	0x0000
Data size 2	High Extra Data Size	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data	SG400, ANSI378 or ISO19794-2 Template	

Example 2: Using extended 6 ~ 22-digit User ID.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x79	0x79
Param1	0x0000	0xFFFF
Param2	0: do not use Wiegand or Relay0 out 1: use Wiegand or Relay0 out	matched score
Data size 1	Low Extra Data Size	Low Extra Data Size
Data size 2	High Extra Data Size	High Extra Data Size
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data	SG400, ANSI378 or ISO19794-2 Template	Extended User ID

CMD_DB_IDENTIFY_EX (0x7A)

- Identify an input fingerprint template within a restricted range of the database.
- Template format of data sent to FMS should be the same format as in SYSINFO.
- SG400 template size is 400 bytes. ANSI378 and ISO19794-2 template size is 800 bytes.
- Wiegand or Relay out is not generated.
- This command does not work with Extended ID.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x7A	0x7A
Param1	First ID number of search	User ID (4 digit)
Param2	Last ID number of search	matched score
Data size 1	Low Extra Data Size	0x0000
Data size 2	High Extra Data Size	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data	SG400, ANSI378 or ISO19794-2 Template	

CMD_INSTANT_VERIFY (0xD0)

- Verify fingerprint by comparing the fingerprint template from external (host) controller and fingerprint template captured from the sensor.
- Param1 is the number of the template in COMMAND packet.

Example 1: SG400 Template

	COMMAND	ACK
Channel	0x00	0x00
Command	0xD0	0xD0
Param1	The number of Template	Matched number of template
Param2	0x0000	0x0000
Data size 1	Low Extra Data Size	0x0000
Data size 2	High Extra Data Size	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data	SG400 Templates	

Example 2: ANSI378 or ISO19794-2 Record

	COMMAND	ACK
Channel	0x00	0x00
Command	0xD0	0xD0
Param1	The record number	Matched finger information1 Upper 8bit: finger position number Lower 8bit: finger view number
Param2	0x0000	Matched finger information2 Upper 8bit: record number Lower 8bit: ANSI378 or ISO19794-2 view number
Data size 1	Low Extra Data Size	0x0000
Data size 2	High Extra Data Size	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data	ANSI378 or ISO19794-2 Records	

CMD_LOAD_MATCH_TEMPLATE (0xFA)

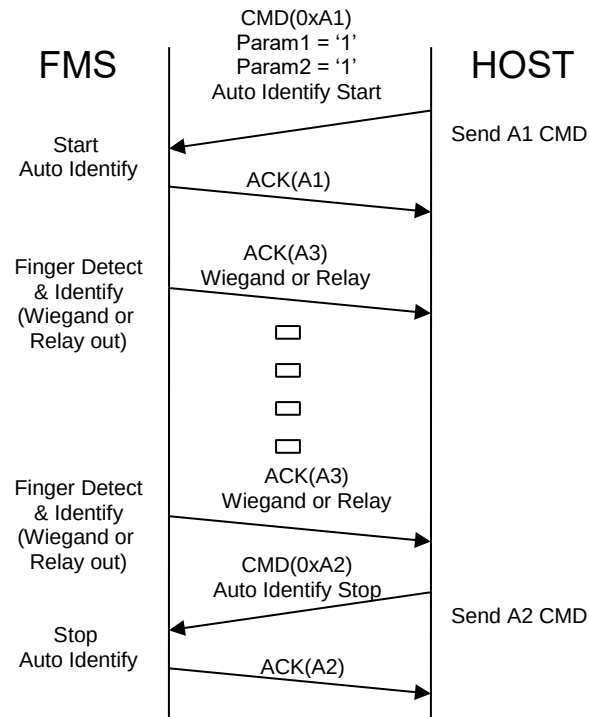
- Verify two fingerprint templates.

	COMMAND	ACK
Channel	0x00	0x00
Command	0xFA	0xFA
Param1	0x0000	Matched score
Param2	0x0000	0x0000
Data size 1	Low Extra Data Size	0x0000
Data size 2	High Extra Data Size	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data	Two SG400 Templates Or TWO ANSI Templates Or TWO ISO Templates	

Auto Identification

The FMS has two modes of automatic identification.

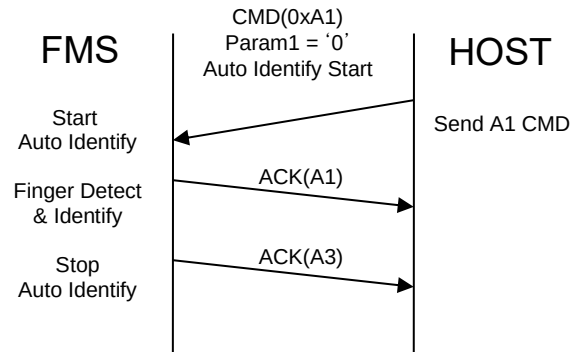
Auto Identify Mode 1



Example 1: Auto Identify Mode 1

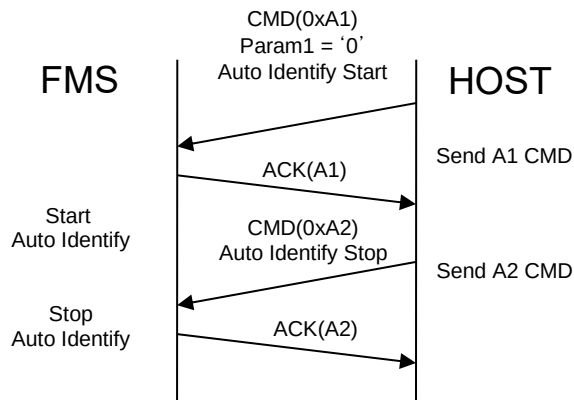
- The FMS Auto Identify function is started by 0xA1 command packet and is stopped by 0xA2 command packet.
- 0xA3 ACK packet is generated whenever a finger is detected.
- Whenever a finger is detected, the FMS identifies the input fingerprint from among the registered fingerprints in the FMS database and sends 0xA3 ACK packet until 0xA2 command packet is received.
- Param2 value is valid when SI_USE_RELAY or SI_WIEGAND_FORMAT of SYSINFO is enabled.
- If Param2 value is '1', a Wiegand signal is generated before the FMS sends ACK packet, and Relay signal is generated after the FMS sends ACK packet.
- If Param2 value is '0', Wiegand and relay signals are not generated.

Auto Identify Mode 2



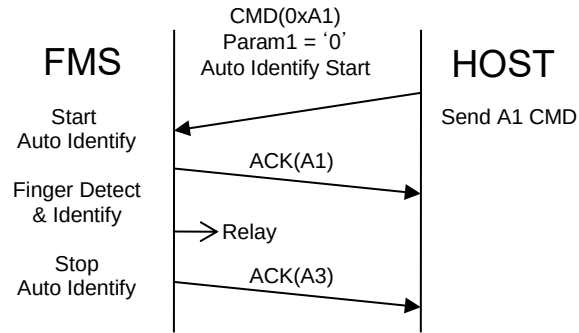
Example 2: Auto Identify Mode 2

- The FMS is operated like the SecuGen FDA02 when Param1 value is '0'.
- The FMS Auto Identify function is started by 0xA1 command packet.
- When a finger is detected, FMS identifies once the input fingerprint from among the registered fingerprints and automatically stops the Auto Identify function.
- 0xA3 ACK packet is generated when a finger is detected when SI_WIEGAND_FORMAT of SYSINFO is disabled.
- If Param2 value is '1', a Wiegand signal is generated before the FMS sends ACK packet, and a relay signal is generated after the FMS sends ACK packet.
- If Param2 value is '0', Wiegand and relay signals are not generated.

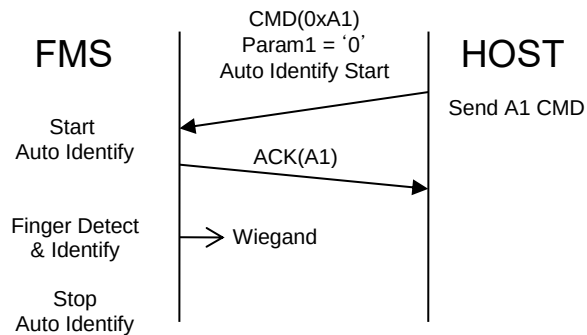


Example 3: Auto Identify Mode 2

- The FMS is operated like FDA02 when Param1 value is '0'.
- The FMS Auto Identify function is started by A1 command packet and is stopped by 0xA2 command packet.

**Example 4: Auto Identify Mode 2**

- The FMS is operated like FDA02 when Param1 value is '0'.
- The FMS Auto Identify function is started by 0xA1 command packet.
- The FMS generates a relay signal when SI_USE_RELAY of SYSINFO is enabled and SI_WIEGAND_FORMAT of SYSINFO is disabled.
- 0xA3 ACK packet is generated after a relay signal is generated.
- When a finger is detected, the FMS generates a relay signal after it identifies the input fingerprint from among the registered fingerprints and stops Auto Identify functions.

**Example 5: Auto Identify Mode 2**

- The FMS is operated like FDA02 when Param1 value is '0'.
- The FMS Auto Identify function is started by 0xA1 command.
- When a finger is detected, the FMS generates a Wiegand signal after it identifies the input fingerprint from among the registered fingerprints and stops Auto Identify functions.
- The FMS generates a Wiegand signal when SI_USE_RELAY of SYSINFO is disabled and SI_WIEGAND_FORMAT of SYSINFO is enabled.
- The FMS does not send 0xA3 ACK packet.

CMD_FP_AUTO_IDENTIFY_START (0xA1)

- When the FMS receives the CMD_FP_AUTO_IDENTIFY command, it continuously compares the input fingerprint from the sensor with all the registered fingerprints in the FMS database until the CMD_FP_AUTO_IDENTIFY_STOP command is received.
- The identify result is sent to the external (host) controller whenever a user places a finger on the sensor.
- When this command is executed, some commands are ignored.
CMD_FP_AUTO_IDENTIFY_STOP, CMD_REQUEST_CONNECT, CMD_SET_SYSTEM_INFO, CMD_GET_SYSTEM_INFO commands are accepted.

	COMMAND	ACK
Channel	0x00	0x00
Command	0xA1	0xA1
Param1	0: operated like FDA02 1: Auto identify function is executed continually.	0x0000
Param2	0: do not use Wiegand or Relay0 out 1: use Wiegand or Relay0 out	0x0000
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

- Whenever a finger is placed on the sensor, the FMS starts matching and returns a result in the ACK packet.

	COMMAND	ACK
Channel	0x00	0x00
Command	0xA3	0xA3
Param1	User ID (4 digit)	0xFFFF
Param2	0x0000 or 0x0001	0x0000 or 0x0001
Data size 1	0x0000	Low Extra Data Size
Data size 2	0x0000	High Extra Data Size
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data		Extended User ID

CMD_FP_AUTO_IDENTIFY_STOP (0xA2)

- Stop Auto Identification mode

	COMMAND	ACK
Channel	0x00	0x00
Command	0xA2	0xA2
Param1	0x0000	0x0000
Param2	0x0000	0x0000
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

Deletion

Delete specified user record or all user records.

Command : CMD_FP_DELETE (0x54)
: CMD_DB_DELETE_REC (0x72)
: CMD_DB_DELETE_ALL (0x76)

CMD_FP_DELETE (0x54)**CMD_DB_DELETE_REC (0x72)**

- Delete a specified user record from the FMS database.

Example 1: Using standard 4-digit User ID.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x54 or 0x72	0x54 or 0x72
Param1	User ID (4 digit)	User ID (4 digit)
Param2	0x00	0x0000
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

Example 2: Using extended 6 ~ 22-digit User ID.

- Must set 0xFFFF in Param1 in command packet

	COMMAND	ACK
Channel	0x00	0x00
Command	0x54 or 0x72	0x54 or 0x72
Param1	0xFFFF	0x0000
Param2	0x0000	0x0000
Data size 1	Low Extra Data Size	0x0000
Data size 2	High Extra Data Size	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data	Extended User ID	

** [Refer to 1.6.1 Extended ID Transfer](#) (Extra Data)

CMD_DB_DELETE_ALL (0x76)

- Delete all user records in the FMS database

	COMMAND	ACK
Channel	0x00	0x00
Command	0x76	0x76
Param1	0x0000	0x0000
Param2	0x0000	0x0000
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

2.3. Reading Fingerprint Image and Template Data

Command : CMD_GET_IMAGE (0x43)
 : CMD_GET_TEMPLATE (0x40)
 : CMD_MAKE_RECORD_START (0x35)
 : CMD_MAKE_RECORD_CONT (0x36)
 : CMD_MAKE_RECORD_END (0x37)

CMD_GET_IMAGE (0x43)

- Read fingerprint RAW image or compressed image in WSQ format from the sensor.
- Image size can be adjusted by Param1 in COMMAND packet.
- When Param2 value is set to '1', the FMS returns most recently captured image.
- **In ISO mode, PARAM1 value must be 0x0001.**

Model	ORG Image size	Half Image size
SDA03P / SDA04P	260*300	130*150
SDA04 PIV	258*332	129*166
Unity 10	252*330	126*165
Unity 20	300*400	150*200

Parameter1		Parameter2	
0x0001	Raw ORG size	0x00XX	
0x0002	Raw Half size	0x00XX	
0x0101	WSQ format, ORG size	0x01XX	BITRATE_5_TO_1
		0x02XX	BITRATE_15_TO_1
0x0102	WSQ format, Half size	0x01XX	BITRATE_5_TO_1
		0x02XX	BITRATE_15_TO_1

- BITRATE_5_TO_1 means 2.25 and BITRATE_15_TO_1 means 0.75

Parameter 2		Description
0xXXXX	0xXX00	Get Image
	0xXX01	Get most recently captured image

	COMMAND	ACK
Channel	0x00	0x00
Command	0x43	0x43
Param1	Parameter1	Parameter1
Param2	Parameter2	Parameter2
Data size 1	0x0000	Low Extra Data Size
Data size 2	0x0000	High Extra Data Size

Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data		SG400 or ANSI378 mode - Raw Fingerprint image - Compressed (WSQ format) Fingerprint image ISO19297-2 mode - ISO fingerprint image format - Header + image

CMD_GET_TEMPLATE (0x40)

- Generate template from the fingerprint image captured on the sensor.
- When template is ANSI378 format, Param1 is finger number in COMMAND packet.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x40	0x40
Param1	0x0000 or number of finger	0x0000 or number of finger
Param2	0x0000	0x0000
Data size 1	0x0000	Low Extra Data Size
Data size 2	0x0000	High Extra Data Size
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data		SG400, ANSI378 or ISO19794-2 Template

CMD_MAKE_RECORD_START (0x35)

- Begin process of generating multiple-view SG400, ANSI378 or ISO19794-2 fingerprint records.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x35	0x35
Param1	Finger number	Finger number
Param2	0x0000	0x0000
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List

Check Sum	Checksum	Checksum
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CMD_MAKE_RECORD_CONT (0x36)

- Capture a fingerprint image and extract template. Accumulate template data in RAM until the CMD_MAKE_RECORD_END command is received.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x36	0x36
Param1	Finger number	Finger number
Param2	0x0000	0x0000
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

CMD_MAKE_RECORD_END (0x37)

- Finalize generation of fingerprint template and download to host controller.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x37	0x37
Param1	0x0000	0x0000
Param2	0x0000	0x0000
Data size 1	0x0000	Low Extra Data Size
Data size 2	0x0000	High Extra Data Size
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data		SG400, ANSI378 or ISO19794-2

2.4. Accessing Fingerprint Database

[Refer to 2.2.1 Structure of Fingerprint Record.](#)

Command : CMD_DB_GET_REC (0x73)
 : CMD_DB_GET_FIRSTREC (0x74)
 : CMD_DB_GET_NEXTREC (0x75)
 : CMD_DB_GET_CURRENTREC (0x77)

CMD_DB_GET_REC (0x73)

- Get a specified user record from the FMS database.
- SG400 template size is 400 bytes. ANSI378 and ISO19794-2 template size is 800 bytes.

Example 1: Using standard 4-digit User ID.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x73	0x73
Param1	User ID (4 digit)	User ID (4 digit)
Param2	0x0000	0x0000
Data size 1	0x0000	Low Extra Data Size
Data size 2	0x0000	High Extra Data Size
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data		User ID (2 bytes) Master Flag (2 bytes) Template1 (400 bytes or 800 bytes) Template2 (400 bytes or 800 bytes) Extended User ID (MAX:11 bytes) Extended User ID Length (1 byte)

Example 2: Using extended 6 ~ 22-digit User ID.

- Must set 0xFFFF in Param1 in command packet

	COMMAND	ACK
Channel	0x00	0x00
Command	0x73	0x73
Param1	0xFFFF	0xFFFF
Param2	0x0000	0x0000
Data size 1	Low Extra Data Size	Low Extra Data Size
Data size 2	High Extra Data Size	High Extra Data Size
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data	Extended User ID	User ID (2 bytes) Master Flag (2 bytes) Template1 (400 bytes or 800 bytes) Template2 (400 bytes or 800 bytes) Extended User ID (MAX:11 bytes) Extended User ID Length (1 byte)

CMD_DB_GET_FIRSTREC (0x74)

- Get the first user record from the FMS database.

CMD_DB_GET_NEXTREC (0x75)

- Get the next user record from the FMS database.

CMD_DB_CURRENTREC (0x77)

- Get the last record obtained by CMD_DB_GET_FIRSTREC or CMD_DB_GET_NEXTREC.
- SG400 template size is 400 bytes. ANSI378 and ISO19794-2 template size is 800 bytes.

Example 1: Using standard 4-digit User ID and extended 6 ~ 22-digit User ID.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x74 or 0x75 or 0x77	0x74 or 0x75 or 0x77
Param1	0x0000	0x0000
Param2	0x0000	0x0000
Data size 1	0x0000	Low Extra Data Size
Data size 2	0x0000	High Extra Data Size
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data	Extended User ID	User ID (2 bytes) Master Flag (2 bytes) Template1 (400 bytes or 800 bytes) Template2 (400 bytes or 800 bytes) Extended User ID (MAX:11 bytes) Extended User ID Length (1 byte)

2.5. Accessing User Data Memory

The FMS provides 16 blocks of user memory. Each block size is 64 Kbytes. A total of 1 Mbytes of user memory is provided. A user can read and write these blocks any time.

Command : CMD_SET_USERDATA (0x92)
 : CMD_GET_USERDATA (0x93)
 : CMD_ERASE_USERDATA (0x94)

CMD_SET_USERDATA (0x92)

- Write user data to memory.
- If user uses 10 Kbytes of memory, the other 54 Kbytes memory are not used.
- Cannot delete user memory to write new data.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x92	0x92
Param1	The number of block	The number of block
Param2	User data size	User data size
Data size 1	Low Extra Data Size	0x0000
Data size 2	High Extra Data Size	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data	User Data (1~64 Kbytes)	

CMD_GET_USERDATA (0x93)

- Read user data from memory.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x93	0x93
Param1	The number of block	The number of block
Param2	User data size	User data size
Data size 1	0x0000	Low Extra Data Size
Data size 2	0x0000	High Extra Data Size
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data		User Data (1~64 Kbytes)

2.6. Other Commands

Command : CMD_REQUEST_CONNECT (0xF0)
 : CMD_GET_VERSION (0x05)
 : CMD_GET_REC_COUNT (0x46)
 : CMD_DB_GET_REC_COUNT (0x70)
 : CMD_GET_MASTER_COUNT (0x47)
 : CMD_IS_REGISTER_USER (0x60)
 : CMD_DB_GET_ID_LIST (0x7D)
 : CMD_GET_MASTER_LIST (0x7E)
 : CMD_RELAY_ON_OFF (0x12)
 : CMD_SET_IO_CONFIG (0xB0)
 : CMD_GET_IO_CONFIG (0xB1)
 : CMD_WRITE_IO (0xB2)
 : CMD_READ_IO (0xB3)
 : CMD_GET_VOID_ID_LIST (0xB6)
 : CMD_SET_WIEGAND_OUT (0x15)
 : CMD_GET_AUTO_ON (0x6F)
 : CMD_GET_MATCH_SCORE (0xF9)
 : CMD_CANCEL (0xA8B8C8D8)

CMD_REQUEST_CONNECT (0xF0)

- Check FMS status to determine if FMS is ready to communicate.
- The host (external controller) should try connecting and wait until it receives ACK with error code 0x00. During DB initialization, the FMS ignores all commands and returns ACK with error code WARNING_DB_LOADING (0x41), which means that it is still initializing the DB.

	COMMAND	ACK
Channel	0x00	0x00
Command	0xF0	0xF0
Param1	0x0000	Version Major Number
Param2	0x0000	Version Minor Number
Data Size 1	0x0000	0x0000
Data Size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List and 1.5 Warning Code List
Check Sum	Checksum	Checksum

Return code	Description
NO_ERROR (0x00)	There was no error during FMS boot up.
ERR_FLASH_OPEN (0x01)	The FMS cannot open flash memory during FMS boot up; need to power reset. If the FMS repeats returning the same error code after power cycle, the flash memory might have crashed or is defective. In this case, the FMS board needs to be replaced.
ERR_FLASH_READ (0x13)	The FMS cannot read flash memory during FMS boot up; need to power reset. If the FMS repeats returning the same error code after power cycle, the flash memory might have crashed or is defective. In this case, the FMS board needs to be replaced.
WARNING_DB_LOADING (0x41)	Returns when the FMS is in the process of loading user data from database during FMS boot up. Must try calling CMD_REQUEST_CONNECT again until no error or warning returned.
WARNING_DB_CHECKSUM (0x42)	Returns when any data is detected as forged or corrupted while loading user data from database during FMS boot up. Recommended to call the CMD_GET_VOID_ID_LIST (0xB6) command to find out which user data have been detected and removed for security reason. Although returning this code, the FMS should work normally for all users except the ones in the void ID list.

CMD_GET_VERSION (0x05)

- Get firmware version.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x05	0x05
Param1	0x0000	Major Version Number
Param2	0x0000	Minor Version Number
Data Size 1	0x0000	0x0000
Data Size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

CMD_GET_REC_COUNT (0x46)

CMD_DB_GET_REC_COUNT (0x70)

- Read the number of registered users.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x46 or 0x70	0x46 0x70
Param1	0x0000	The number of registered users
Param2	0x0000	The number of remaining available users
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

CMD_DB_GET_MASTER_COUNT (0x47)

- Read the number of registered masters.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x47	0x47
Param1	0x0000	The number of registered masters
Param2	0x0000	0x0000
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

CMD_IS_REGISTERED_USER (0x60)

- Check whether a specified user is registered
- Using standard 4-digit User ID

	COMMAND	ACK
Channel	0x00	0x00
Command	0x60	0x60
Param1	User ID (4 digit)	User ID (4 digit)
Param2	0x0000	Master Flag
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

- Using extended 6 ~ 22-digit User ID.
- Must set 0xFFFF in Param1 in command packet.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x60	0x60
Param1	0xFFFF	0xFFFF
Param2	0x0000	Master Flag
Data size 1	Low Extra Data Size	0x0000
Data size 2	High Extra Data Size	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum
Extra Data	Extended User ID	

** [Refer to 1.6.1 Extended ID Transfer](#) (Extra Data)

CMD_GET_VOID_LIST (0xB6)

- This command retrieves the list of User IDs that have been detected as forged or corrupt and removed from the database when WARNING_DB_CHECKSUM is returned by calling CMD_REQUEST_CONNECT (0xF0).
- The FMS returns a maximum of 50 in the list of crashed User IDs.
- Packet (COMMAND, ACK) structure is the same as CMD_DB_GET_ID_LIST command.

CMD_DB_GET_ID_LIST (0x7D)

CMD_DB_GET_MASTER_LIST (0x7E)

- Get a list of all registered User IDs or Master IDs.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x7D or 0x7E	0x7D or 0x7E
Param1	0: operated like FDA02 1: operated as below table	The number of registered users or master users
Param2	0x0000	0x0000
Data size 1	0x0000	Low Extra Data Size
Data size 2	0x0000	High Extra Data Size
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum ID (11 bytes) + ID Length (1 byte) + ID (11 bytes) + ID Length (1 byte) ++ ID (11 bytes) + ID Length (1 byte)

- **Example:** When returning 4 User IDs in the master list (1234567890123456789012, 1234, 5678, 987654)
- **ACK =>** Param1: 0x04, Data size 1: 0x0030

1	2	3	4	5	6	7	8	9	10	11	12
1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte
0x12	0x34	0x56	0x78	0x90	0x12	0x34	0x56	0x78	0x90	0x12	0x0B

13	14	15	16	17	18	19	20	21	22	23	24
1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte
0x34	0x12	0xFF	0xFF	0xFF	0xFF	0xFF	0xFF	0xFF	0xFF	0xFF	0x02

25	26	27	28	29	30	31	32	33	34	35	36
1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte
0x78	0x56	0xFF	0xFF	0xFF	0xFF	0xFF	0xFF	0xFF	0xFF	0xFF	0x02

37	38	39	40	41	42	43	44	45	46	47	48
1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte	1 byte
0x98	0x76	0x54	0xFF	0xFF	0xFF	0xFF	0xFF	0xFF	0xFF	0xFF	0x03

CMD_RELAY_ON_OFF (0x12)

- Control OUT signals of Relay0 or Relay1.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x12	0x12
Param1	0: RELAY0 1: RELAY1	0: RELAY0 1: RELAY1
Param2	0: OFF 1: ON	0: OFF 1: ON
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

CMD_SET_IO_CONFIG (0xB0)

- Configure IO.

	COMMAND	ACK
Channel	0x00	0x00
Command	0xB0	0xB0
Param1	Parameter1	Parameter1
Param2	0x0000	0x0000
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

Parameter1					
	IO4	IO3	IO2	IO1	IO0
BIT32 ~ BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
0x0	0: input, 1: output				

** IO0, IO1, IO2 are connected to internal pull-up resistor (10 Kohm)

CMD_GET_IO_CONFIG (0xB1)

- Get IO configuration.

	COMMAND	ACK
Channel	0x00	0x00
Command	0xB1	0xB1
Param1	0x0000	Return IO configuration
Param2	0x0000	0x0000
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

CMD_WRITE_IO (0xB2)

- Set OUT signals of IO.

	COMMAND	ACK
Channel	0x00	0x00
Command	0xB2	0xB2
Param1	GPIO value	GPIO value
Param2	0x0000	0x0000
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

CMD_READ_IO (0xB3)

- Get IN signals of IO.

	COMMAND	ACK
Channel	0x00	0x00
Command	0xB3	0xB3
Param1	0x0000	GPIO value
Param2	0x0000	0x0000
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

CMD_GET_MATCH_SCORE (0xF9)

- Get last matching score and user ID. (User ID only valid 4 digits)

	COMMAND	ACK
Channel	0x00	0x00
Command	0xF9	0xF9
Param1	0x0000	Matched score
Param2	0x0000	Matched user ID (4 digits)
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

CMD_SET_WIEGAND_OUT (0x15)

- Output user ID to Wiegand port.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x15	0x15
Param1	User ID (4 digit)	User ID (4 digits)
Param2	0x0000	0x0000
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

CMD_GET_AUTO_ON (0x6F)

- Get Auto On status of sensor module.

	COMMAND	ACK
Channel	0x00	0x00
Command	0x6F	0x6F
Param1	0x0000	0: finger not detected on the sensor 1: finger detected on the sensor
Param2	0x0000	0x0000
Data size 1	0x0000	0x0000
Data size 2	0x0000	0x0000
Error Code	0x00	Refer to 1.4 Error Code List
Check Sum	Checksum	Checksum

CMD_CANCEL (0xA8B8C8D8)

- This command is used to stop image capture.
- If the device receives a CMD_CANCEL command while the device is capturing images, it will stop capturing images.
- To stop capturing images, send 0xA8, 0xB8, 0xC8, 0xD8 to the FMS device sequentially.
- The CMD_CANCEL command is valid after the host sends a command that includes capture commands such as VERIFY, IDENTIFY, INSTANT VERIFY.

